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Nanomaterial-Enhanced Organic-Rich Shales: Data-Driven Insights into Wettability Predictions for Underground Hydrogen Storage Design

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Abstract

Shale caprocks, often organic-rich, play a significant role in hydrogen sequestration, but their wettability is typically altered from water-wet to H_2 -wet due to the organic matter, compromising their sealing performance. Silica nanofluids can restore water-wet conditions, which is vital for effective containment of H_2 in geological formations. However, laboratory experiments to test the effects of nanomaterials on wettability and interfacial tension (IFT) are resource-intensive and time-consuming. To address this challenge, data-driven machine learning (ML) models offer a promising alternative, allowing for the prediction of key UHS parameters—such as advancing and receding contact angles without the need for extensive laboratory testing. We have applied multiple ML models, including gradient boosting, extreme gradient boosting (XGboost), Catboost, random forest, and extra trees to predict the wettability of organic rich shale samples treated with silica nanofluids. The ML models were trained on a comprehensive dataset generated through laboratory experiments conducted under a wide range of pressure (0.5 – 1600psi) and temperature (298-323 K) conditions, using brine (1 wt.% KCl + 2 wt.% NaCl) and varying silica nanofluid concentrations (0.05–1.0 wt.%). Advanced visualization techniques, including swarm maps, box plots, and heatmap plots, were utilized to analyze the experimental dataset thoroughly. To enhance the generalization capabilities of the ML models, we employed Bayesian optimization for hyperparameter tuning, along with k-fold cross-validation and grid-search optimization techniques. Both statistical and graphical analyses were performed to show the reliability and performance of the models. The results demonstrated that the implemented ML models, particularly the XGboost and Catboost, accurately predicted the wettability behavior under various operating conditions. The training and testing average R^2 scores were above 0.998 and 0.949, respectively, confirming the high accuracy of the predictions. These findings highlight the reliability of the ML algorithms in predicting the complex interaction between shale, brine, and H_2 . Further investigation into the feature importance revealed that nanomaterials had the greatest influence on the wettability of the ternary system. The silica nanofluid concentrations played a key role in modifying wettability, with varying impacts on the receding contact angles, emphasizing the effect of nanomaterial

treatment. The model's predictions were consistent with experimental results, suggesting that geological conditions, including pressure and temperature, have a profound influence on rock mineral wettability, directly impacting H₂ geo-storage capacity. This study is the first to employ data-driven ML models to predict critical UHS parameters in organic-rich shales treated with silica nanofluids. The ability of the ML models to predict UHS parameters across a wide range of conditions provides a valuable tool for optimizing UHS system designs, offering a more efficient and scalable approach compared to traditional experimental methods.

Introduction

Underground hydrogen storage (UHS) is a cornerstone of the hydrogen economy, providing a means to store large volumes of H₂ to balance supply and demand fluctuations (Ali et al., 2025; Raza et al., 2024; Tariq et al., 2025b). Geological formations such as depleted gas reservoirs, saline aquifers, and salt caverns have been extensively studied for their storage potential due to their large capacities and ability to maintain containment over extended periods (Hematpur et al., 2023; Miocic et al., 2023; Muhammed et al., 2022; Sambo et al., 2022; Tariq et al., 2025a). Organic-rich shales, characterized by low porosity, low permeability, and high organic content, have gained attention as both storage media and caprocks. Their nanoscale pore networks, and adsorption capacity make them suitable for immobile H₂ storage, with adsorbed gas constituting a significant portion of the total stored volume (Hosseini et al., 2022; Tariq et al., 2023c). Studies highlight that shales can store 20–85% of gas in an adsorbed phase, reducing leakage risks compared to conventional reservoirs. However, the success of UHS depends on understanding critical interfacial properties, including wettability and interfacial tension (IFT) between hydrogen (H₂), brine, and reservoir rocks, which govern storage efficiency and containment security (Ali et al., 2022, 2021; Muhammed et al., 2023). These properties are particularly complex in organic-rich shale caprocks, where high total organic content (TOC) can alter wettability, potentially compromising sealing capacity.

The efficacy of UHS relies on several trapping mechanisms. Structural trapping involves impermeable caprocks, such as shales, that act as barriers to H₂ migration. Residual trapping occurs when capillary forces in hydrophilic pore systems immobilize H₂ within the rock matrix. Solubility trapping, where H₂ dissolves into formation fluids, is less significant due to H₂'s low solubility. Unlike CO₂ sequestration, mineral trapping—where gas reacts with minerals to form stable compounds—is undesirable for UHS, as it hinders H₂ extraction during demand cycles. The interplay between wettability and IFT is critical to these mechanisms, as hydrophilic conditions enhance capillary sealing and containment security. Wettability governs the interaction between rock surfaces, brine, and H₂, influencing capillary pressure and gas mobility. Strongly water-wet conditions are optimal for maximizing capillary trapping and minimizing leakage risks. IFT between H₂ and brine affects interface stability and capillary forces, directly impacting storage efficiency. Recent studies have investigated H₂ wettability in various rock systems, including sandstones, carbonates, and shales. Capillary entry pressure and maximum H₂ column height, calculated using models like the Laplace-Young equation, are key metrics for assessing sealing efficiency in caprocks. High-TOC shales, with their organic-rich composition, often exhibit hydrophobic tendencies, necessitating wettability modification to enhance storage performance (Usman et al., 2025).

Nanofluids, suspensions of nanoparticles in base fluids, have emerged as a transformative tool for modifying wettability and IFT in geological storage systems. Silica-based nanofluids have demonstrated the ability to shift hydrophobic rock surfaces to hydrophilic states, enhancing capillary trapping and structural containment. Research on CO₂/brine/rock systems has shown that nanoparticle concentration, base fluid composition, and reservoir conditions (e.g., pressure and temperature) are critical parameters for optimizing nanofluid performance. However, the application of nanofluids to H₂ storage remains limited, with few studies exploring their impact on H₂/brine/shale systems. Preliminary findings suggest that silica nanofluids

can improve wettability under high-pressure, high-temperature conditions, but comprehensive studies on organic-rich shales are lacking.

Organic-rich shales offer unique advantages for H₂ storage, including enhanced adsorption due to their organic content and nanoscale pores. The adsorption capacity is influenced by organic matter type, maturity, and inorganic mineral composition. Shales also facilitate additional benefits, such as increased methane recovery during H₂ injection and reduced leakage risks due to their low permeability. However, the hydrophobic nature of high-TOC shales poses challenges for capillary trapping, underscoring the need for wettability alteration strategies. Studies on proxy organic acids and pure minerals have provided insights into the role of TOC in capillary trapping, but real-world shale samples under reservoir conditions require further investigation.

While significant progress has been made in understanding UHS and nanofluid applications, several knowledge gaps persist. The wettability behavior of H₂ in organic-rich shales under geological conditions is not fully characterized, particularly with respect to silica nanofluid treatments. The impact of varying nanoparticle concentrations on H₂/shale IFT and sealing efficiency remains underexplored. Furthermore, the interplay between shale mineralogy, organic content, and nanofluid performance needs systematic evaluation to develop scalable solutions for UHS. Addressing these gaps is critical to advancing the deployment of shales as a reliable storage medium for a hydrogen-based economy.

Laboratory experiments under simulated reservoir conditions provide the most accurate measurements of wettability and IFT but are time-consuming, costly, and challenging, especially when evaluating diverse pressure, temperature, and fluid composition scenarios (Wang et al., 2023). To overcome these limitations, data-driven machine learning (ML) models offer an efficient and scalable alternative for predicting key UHS parameters, such as advancing and receding contact angles and IFT (Gao et al., 2024; Vo Thanh et al., 2024; Zhao et al., 2024). By leveraging comprehensive experimental datasets, ML models can capture complex, nonlinear relationships between input parameters (e.g., pressure and nanofluid treatments) and output properties, enabling rapid assessment of storage feasibility. This study develops and evaluates multiple ML models, including decision trees, random forests, gradient boosting, adaptive gradient boosting, and extreme gradient boosting, to predict wettability and IFT in high-TOC Jordanian shale samples treated with silica nanofluids. The models are trained on experimental data spanning a wide range of pressures (0.5–1600 psi), temperatures (323 K), and nanofluid concentrations (0.05–1.0 wt.%). The findings aim to provide a robust, predictive framework for optimizing UHS design, enhancing storage capacity, and ensuring long-term containment security in large-scale hydrogen storage projects.

Methodology

Exploratory Data Analysis

A comprehensive experimental data was collected from our previous study (Alanazi et al., 2025) in order to train the machine learning (ML) models. The dataset covers a realistic pressure range of 50 – 1600 psi. The experiments were conducted on four different types of organically rich shales enhanced with nanofluids. The XRD of these shale samples, along with their corresponding total organic carbon (TOC) levels are shown in Figure 1. To streamline the machine learning process, minor mineral constituents such as Berlinite, Apatite, and Langite were grouped under a single category labelled as "Tracers." The XRD data indicate that the primary mineral phases present in the samples are Calcite, Quartz, and Tracers, with the latter representing the combined contribution of these less abundant components.

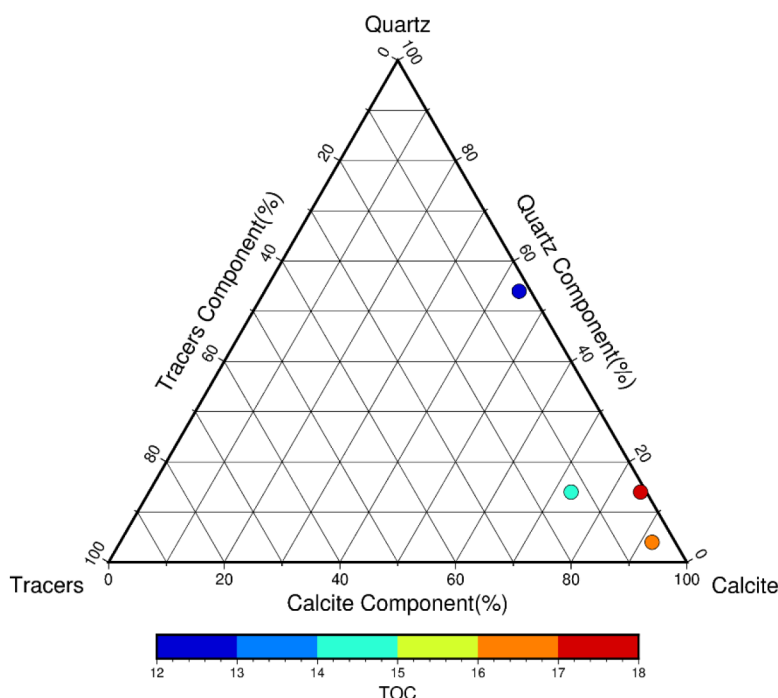


Figure 1—The ternary diagram represents the XRD data of the organic rich rock samples used in this study. The data is taken from (Alanazi et al., 2025).

Furthermore, exploratory statistical analysis of the experimentally measured dataset is summarized in Table 1. The dataset includes several parameters, such as the XRD of various minerals (calcite, quartz, apatite), TOC, roughness, pore volume, and average pore size, in addition to contact angles (θ_R for receding and θ_A for advancing contact angles). The contact angles range from 10° to 106° , with a mean value of 40.77° for advancing contact angles and 33.77° for receding contact angles. The skewness values for the contact angles indicate a moderate positive skew, suggesting that most of the contact angle values are concentrated at the lower end of the range, with a relatively small number of extreme values. The kurtosis values further support this, indicating a slight peaked distribution. The standard deviations (16.92 for receding and 18.59 for advancing contact angles) and variances (286.40 for receding and 345.48 for advancing contact angles) highlight the variability in the contact angle measurements. The pressure variable shows low skewness and a high variance, indicating significant variability in pressure across the dataset, which could have a substantial impact on wettability predictions. The ML models were trained using input parameters derived from the experimental dataset, such as pressure, temperature, and the compositions of H_2 , and brine, alongside the wettability characteristics. This analysis provides critical insights into the distribution and variation in the experimental dataset, which are key factors influencing the model's performance and accuracy in predicting contact angles.

Figure 2 presents the feature importance scores for the input parameters used in the model, highlighting the relationship between the various features and the predicted contact angles. The analysis is based on Pearson's, Spearman's, and Kendall's correlation coefficients, providing a clear understanding of which factors (such as pressure, temperature, or concentration) have the most significant impact on wettability predictions.

Table 1—Statistical analysis of the experimentally measured dataset for contact angle prediction.

Parameters	Mean	Min	Max	25%	50%	75%	Skewness	Kurtosis	Variance	Std
Calcite (v/v)	73.50	44.00	92.00	65.75	79.00	86.75	-0.73	-1.02	338.67	18.40
Quartz (v/v)	21.50	4.00	54.00	11.50	14.00	24.00	1.00	-0.75	371.40	19.27
Tracers (v/v)	4.35	0.00	13.00	1.05	2.70	6.00	0.86	-0.92	21.72	4.66
TOC (wt.%)	15.48	12.93	17.94	14.40	15.52	16.60	-0.06	-1.22	3.36	1.83
Surface Roughness (nm)	191.50	89.00	399.00	124.25	139.00	206.25	1.06	-0.72	14879.53	121.98
Pore volume (cm ³ /g)	0.02	0.01	0.03	0.02	0.03	0.03	-0.76	-1.06	0.00	0.00
Avg. pore size Diameter (nm)	29.50	14.73	41.42	24.11	30.92	36.31	-0.37	-1.21	98.54	9.93
Nano SiO ₂ (wt.%)	0.36	0.00	1.00	0.05	0.20	0.80	0.72	-1.09	0.13	0.36
Pressure (psi)	810	50	1600	400	800	1200	0	-1	306590	554
ρ G	4.01	0.26	7.79	2.04	4.02	5.93	0.01	-1.34	7.24	2.69
Receding Contact Angle θ_R (°)	33.77	10.00	94.00	23.00	31.00	38.00	1.60	2.44	286.40	16.92
Advancing Contact Angle θ_A (°)	40.77	16.00	106.00	30.00	36.00	44.00	1.72	2.65	345.48	18.59

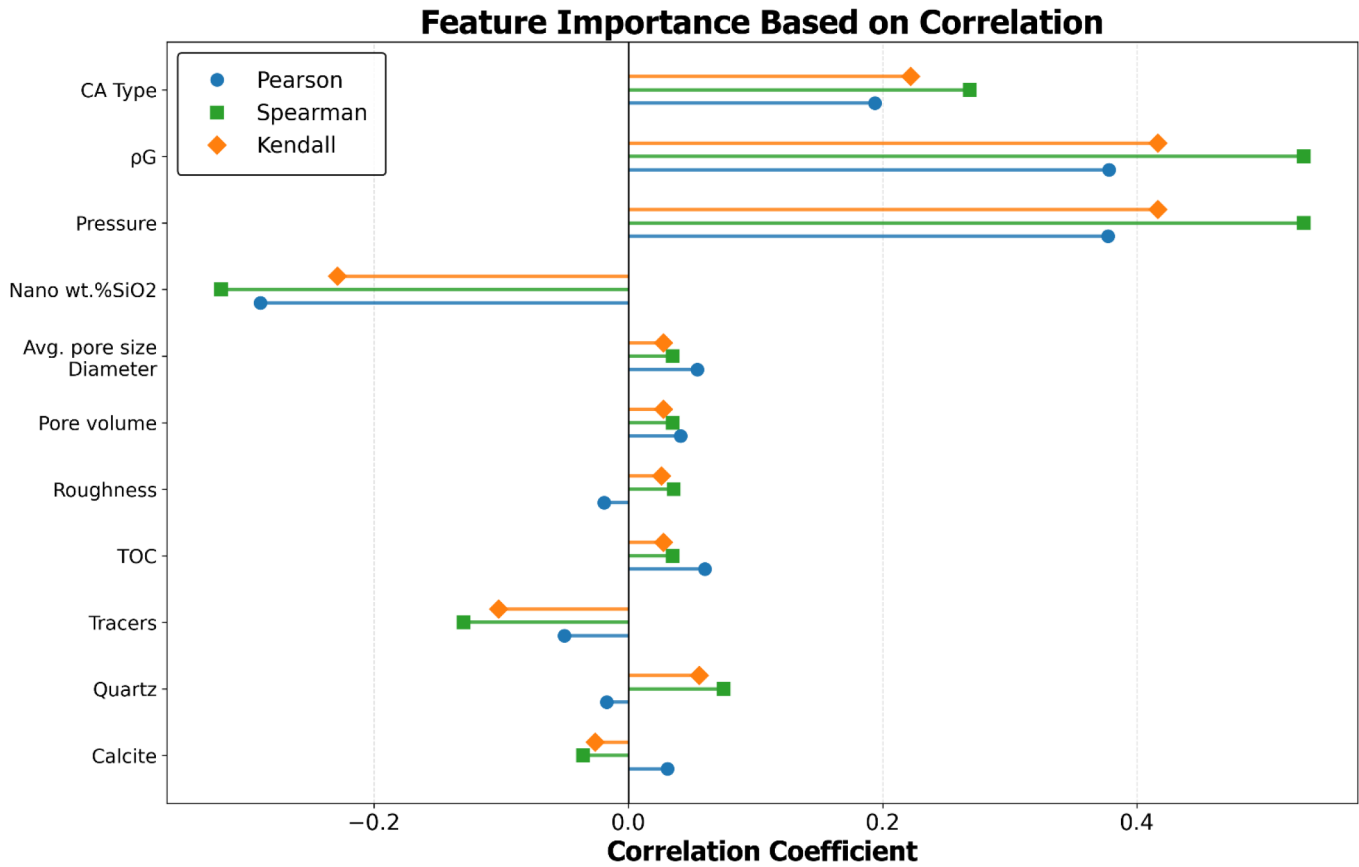


Figure 2—The feature importance of the input parameters with contact angles using Pearson's, Spearman's, and Kendall's criteria.

Figure 3 shows the Swarm distributions of the input features, displayed in box plots. These plots help visualize the distribution and variability of key input parameters, such as pressure, temperature, and organic or nano-aged concentration, across the dataset. These plots allow for better understanding of the feature distributions and their impact on the contact angle predictions.

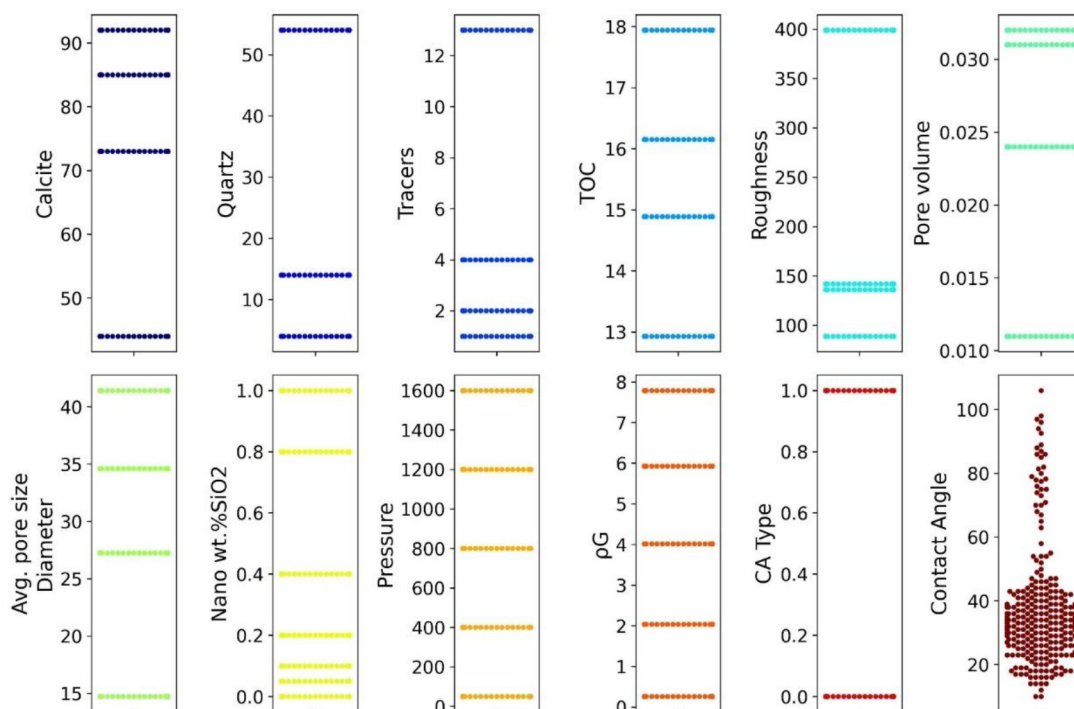


Figure 3—Swarm distributions of input feature shown in box plots.

Machine Learning Techniques

The machine learning techniques used in this study represent a diverse set of methods that have proven effective in predictive modeling tasks (Mustafa et al., 2023; Tariq et al., 2021). Machine learning has been successfully used for the prediction of wettability of CO₂ and Hydrogen geo-storage (Tariq et al., 2023a, 2023b). Random Forest, a popular ensemble learning method, builds multiple decision trees during training and merges their outputs to improve prediction accuracy and control overfitting. This technique works well for complex datasets with non-linear relationships and is highly interpretable, as it provides insights into the importance of input features (Tembely et al., 2020). XGBoost, another ensemble method, utilizes gradient boosting to build strong predictive models by combining the outputs of multiple weak learners. It is known for its high efficiency, scalability, and performance, especially in handling large datasets and avoiding overfitting (Othman et al., 2023). Gradient Boosting, like XGBoost, also builds models sequentially by correcting the errors made by previous models, making it highly effective in capturing intricate patterns in data. CatBoost is a gradient boosting algorithm specifically designed to handle categorical features efficiently, making it particularly suitable for datasets with mixed types of data. It performs well without requiring extensive data preprocessing, such as one-hot encoding, and is known for its robustness and ability to minimize overfitting (Hancock and Khoshgoftaar, 2020). Extra Trees, or Extremely Randomized Trees, is an ensemble method that builds many trees and selects random splits, which helps to reduce variance and improve model stability. This technique tends to work well when there is a large number of features or high-dimensional data (Alsariera et al., 2020; Mastelini et al., 2023; Pagliaro, 2023). Finally, Bayesian Ridge is a regression technique that applies a probabilistic approach to model the relationship between the input features and the target variable (Almutiri et al., 2023; Mostafa et al., 2020; Yang and Yang, 2020). It assumes that the weights in the linear regression model follow Gaussian distribution and use Bayesian inference to estimate the model parameters. This method is particularly useful for scenarios where uncertainty in predictions is important and is capable of providing robust predictions even with noisy data (Boujelbene et al., 2023; Imane et al., 2022). Together, these techniques offer a comprehensive suite of tools for accurately predicting the wettability behavior of the mineral/H₂/brine system under various conditions.

Performance Metrics

To assess the prediction abilities of ML models, various evaluation criteria were utilized. These criteria encompassed scatter plots, average absolute percentage error (MAPE), root mean square error (RMSE), and coefficient of determination (R^2), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Standard Deviation (Std), Variance, Maximum Error (E_{max}), and Minimum Error (E_{min}). The definition of these metrics is given in Eqs (1 – 8).

$$R^2 = \left(\frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \right)^2 \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n_{samples}} \sum_{i=1}^{n_{samples}} (x_{measured} - x_{pred})^2} \quad (2)$$

$$MAE = \frac{1}{n_{samples}} \sum |x_{measured} - x_{pred}| \quad (3)$$

$$MAPE = \frac{1}{n_{samples}} \sum |x_{measured} - x_{pred}| \times \frac{100}{x_{exp}} \quad (4)$$

$$SD = \sqrt{\frac{1}{n_{samples} - 1} \sum_{i=1}^n \left(\frac{x_{measured} - x_{pred}}{x_{measured}} \right)^2} \quad (5)$$

$$Variance = \frac{1}{n_{samples} - 1} \sum_{i=1}^n \left(\frac{x_{measured} - x_{pred}}{x_{measured}} \right)^2 \quad (6)$$

$$E_{max} = \max(|x_{measured} - x_{pred}|) \quad (7)$$

$$E_{min} = \min(|x_{measured} - x_{pred}|) \quad (8)$$

Workflow Methodology

The workflow methodology (shown in Figure 4) for this study involves a structured approach to model the wettability behavior of the nano material enhanced organically rich shale/H₂/brine system under varying experimental conditions. Initially, raw experimental data is collected from contact angle measurements conducted on organic rich shales from our previous study (Alanazi et al., 2025). The data includes varying pressure (0.1 – 20 MPa) and temperature (298 – 323 K) conditions, along with different organic and nanofluid treatments. The dataset is preprocessed and organized for machine learning model training, where exploratory data analysis (EDA) techniques were applied to identify important features such as pressure, temperature, organic and nano-aged concentrations, and contact angle values. Following data preparation, multiple machine learning models—including Extra Trees, CatBoost, Gradient Boosting, XGBoost, and Random Forest—were used to predict the contact angles for the organically rich shale/H₂/brine system. Each model was trained using the preprocessed dataset, with pressure, temperature, organic and nanofluid concentrations serving as input parameters. The model development process incorporates k-fold cross-validation and grid search optimization to enhance generalization and fine-tune hyperparameters. Once the models were trained, they were evaluated using performance metrics such as R^2 , RMSE, MAE, and MAPE. These metrics assess the accuracy and reliability of the models in predicting contact angles under both training and testing conditions. Feature importance was performed using correlation coefficients (Pearson's, Spearman's, and Kendall's) to identify key variables influencing the wettability prediction. Finally, sensitivity analysis was conducted to evaluate the impact of key parameters—such as pressure, and nanofluid concentration—on the wettability behavior of the system. The predicted values were compared with experimental data to assess the accuracy and reliability of the model's predictions. This methodology

allows for a comprehensive understanding of the factors influencing wettability in the context of hydrogen geo-storage.

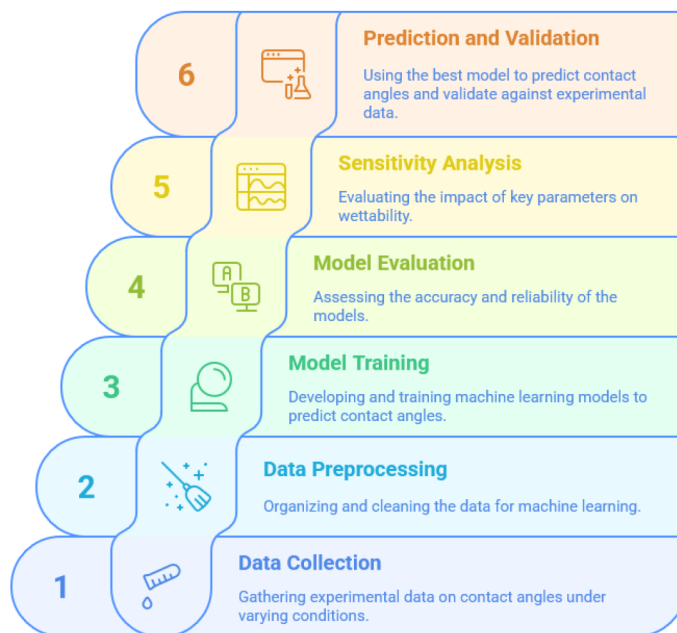


Figure 4—Workflow methodology for predicting the wettability behavior of the nana material enhanced organic rich shales/H₂/brine system using machine learning models.

Results and Discussions

The results from the various machine learning models reveal distinct patterns in terms of both training and testing performance, highlighting the strengths and limitations of each algorithm in predicting wettability behavior. The detailed results are present in [Table 2](#). Starting with the R^2 values, XGBoost and CatBoost models show exceptionally high R^2 values on the training data, indicating a near-perfect fit (R^2 of 0.998 for XGBoost and 0.998 for CatBoost). However, their performance on the test set, with R^2 values of 0.949 for both models, suggests some overfitting during the training phase. In contrast, Random Forest and Gradient Boosting exhibit more balanced performance, with R^2 values of 0.990 and 0.982 on the training data, and 0.889 and 0.915 on the testing data, respectively. Extra Trees, with an R^2 of 0.987 on the training data and 0.902 on the test data, demonstrates strong performance but still shows a slight decrease in performance on the test set. The Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) values further reinforce these observations. XGBoost, with an RMSE of 0.835 on the training data and 3.750 on the test data, shows superior accuracy during training but experiences some loss of generalization on the test set. Extra Trees exhibits an RMSE of 2.085 on the training data and 5.185 on the test data, indicating an overfitting problem where the model memorizes the training data but struggles with new data. The Mean Absolute Percentage Error (MAPE) results highlight that XGBoost and CatBoost exhibit the lowest MAPE values (2.053 for training and 9.080 for testing for XGBoost, and 1.769 for training and 8.238 for testing for CatBoost), indicating more accurate predictions compared to other models, such as Extra Trees (MAPE of 4.903 for training and 14.542 for testing). The variance analysis shows that XGBoost and CatBoost have lower variance on both the training and testing data, which is indicative of their stability across different data splits. In contrast, Random Forest and Extra Trees show relatively higher variance, particularly on the test set, suggesting a less reliable generalization. Overall, XGBoost and CatBoost provide high predictive performance but exhibit slight overfitting, while Random Forest and Gradient Boosting offer more stable results across both training and testing datasets. The Extra Trees model, although performing well on the training set, suffers from significant overfitting, making it less reliable for unseen data. Based on these

results, XGBoost and CatBoost are the most promising models for predicting wettability behavior in the given system. Figure 5 illustrates the parity crossplots of the training and testing datasets, showing the predicted contact angles versus actual measured values, confirming the models' accuracy. Figure 6 presents the Kernel Density Estimation (KDE) plots of the residual errors, providing insights into the consistency and reliability of the models' predictions. Figure 7 shows the feature importances obtained from the trained ML models. Figure 7 confirmed that the impact of Nanofluid concentrations is the most significant on the contact angles.

Table 2—Performance metrics of various machine learning models for predicting the contact angles of the mineral/H₂/brine system. Metrics include R², RMSE, MAE, MAPE, and statistical measures on both training and testing

Model	Random Forest	XGBoost	Gradient Boosting	CatBoost	Extra Trees
R ² Train	0.990	0.998	0.982	0.998	0.987
R ² Test	0.889	0.949	0.915	0.949	0.902
RMSE Train	1.809	0.835	2.434	0.714	2.085
RMSE Test	5.531	3.750	4.834	3.753	5.185
MAE Train	1.424	0.645	1.918	0.557	1.595
MAE Test	4.217	2.574	3.327	2.545	3.864
MAPE Train	4.510	2.053	6.035	1.769	4.903
MAPE Test	15.942	9.080	11.680	8.238	14.542
Std Train	1.808	0.835	2.434	0.714	2.085
Std Test	5.422	3.749	4.777	3.732	5.109
Variance Train	3.269	0.697	5.927	0.509	4.346
Variance Test	29.399	14.058	22.820	13.930	26.102
E _{max} Train	6.672	2.849	7.300	2.662	8.454
E _{max} Test	19.420	14.812	19.100	13.274	15.995
E _{min} Train	0.010	0.003	0.009	0.001	0.006
E _{min} Test	0.040	0.224	0.087	0.088	0.091

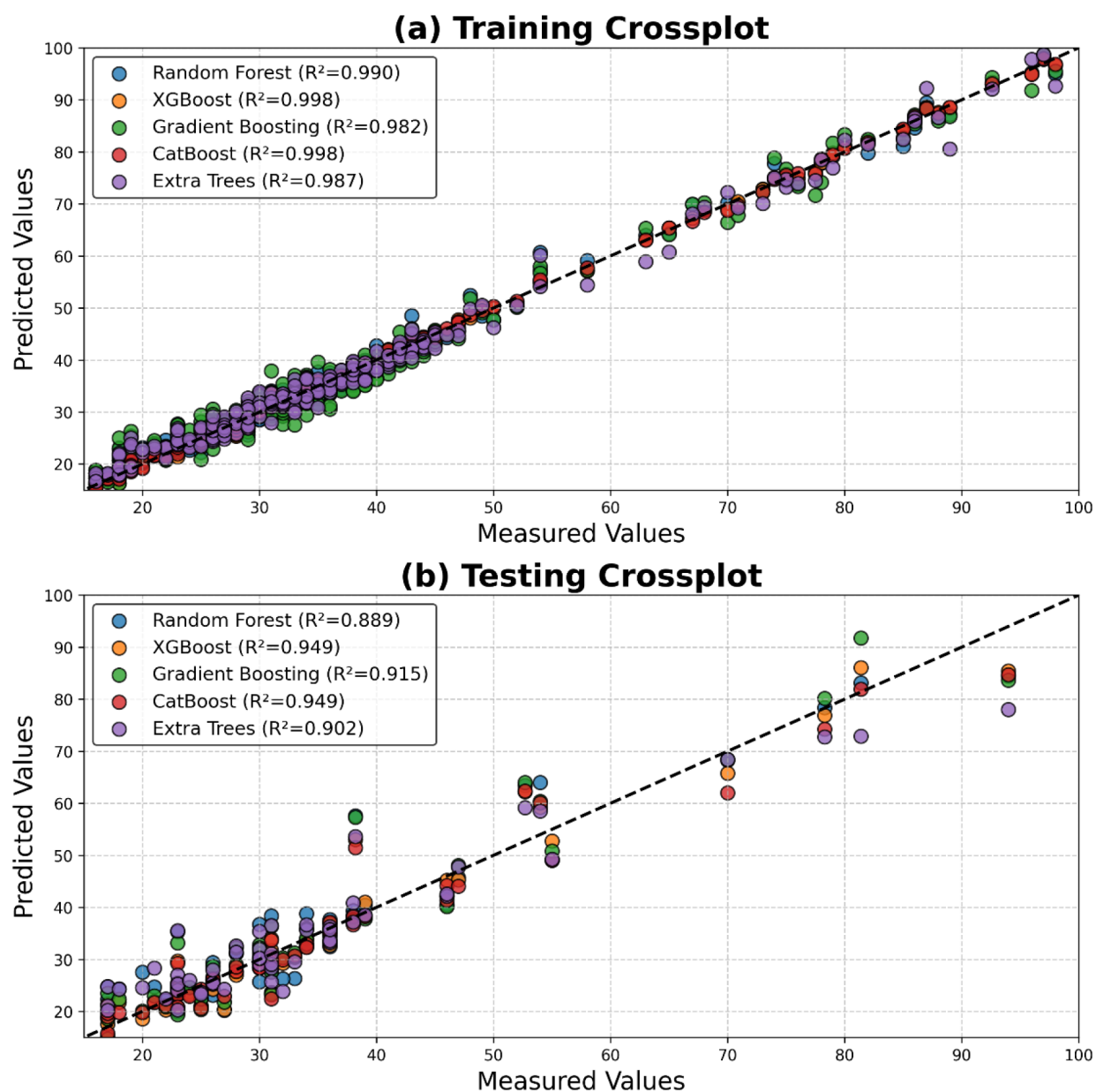
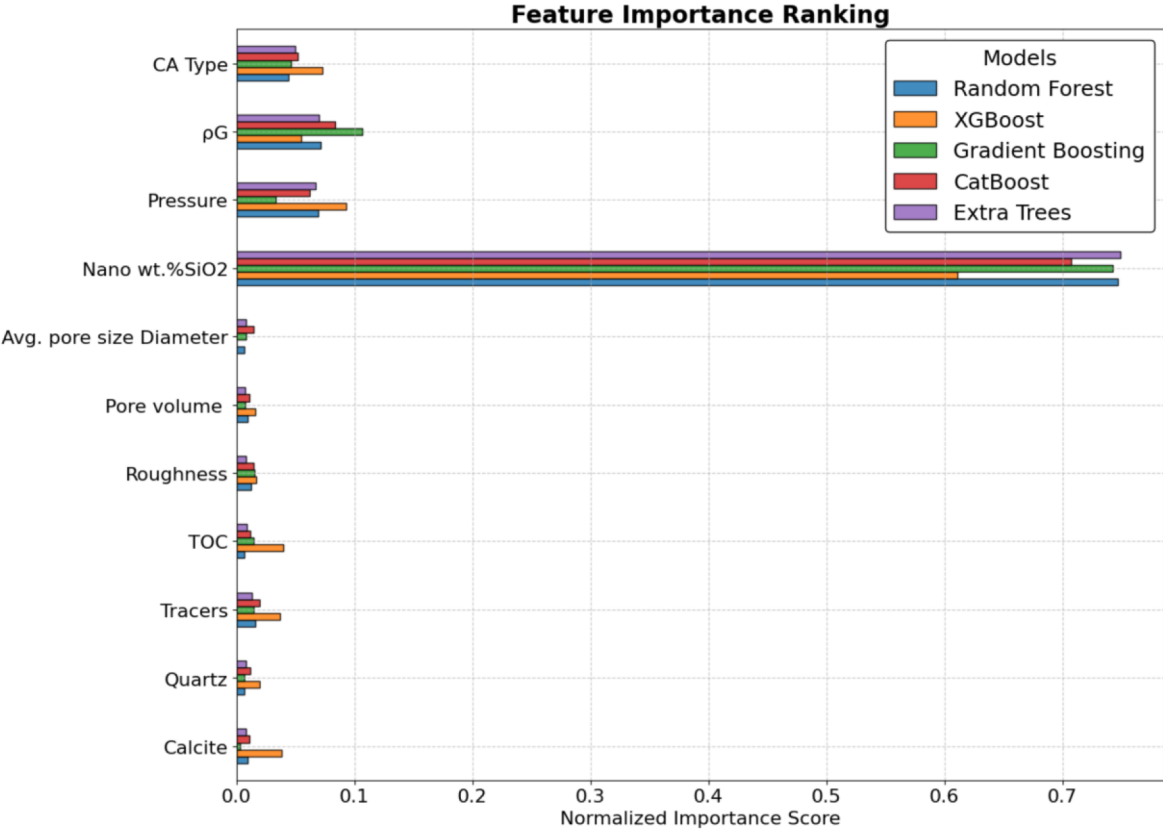
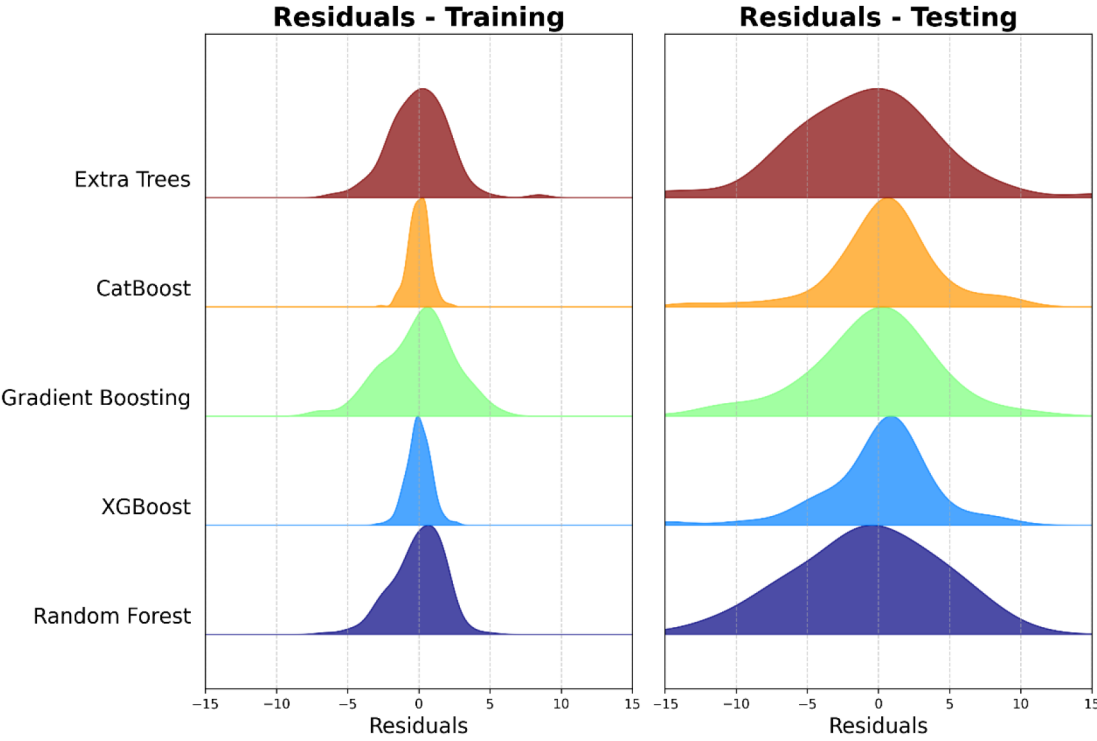


Figure 5—Parity crossplots of training and testing for the contact angle predictions



The trained machine learning model—XGBoost, identified as the best-performing algorithm in this study—was utilized to predict both advancing and receding contact angles under conditions where experimental measurements were unavailable. Figure 8 illustrates the model's predictions at a constant temperature of 323 K, across a pressure range of 14.5 to 1600 psi, and for four different silica nanofluid concentrations (0.25, 0.5, 0.75, and 1.0 wt.%) applied to four types of organic-rich shales (JO-1, JO-2, JO-3, and JO-4). The results show that increasing nanofluid concentrations led to a reversal of wettability toward a water-wet state. Among the tested concentrations, 0.5 wt.% exhibited the most optimal wettability alteration, suggesting a nonlinear relationship between concentration and contact angle response. Additionally, an increase in pressure consistently resulted in higher contact angle values across all shale samples. This behavior is attributed to the enhanced intermolecular interactions between gas molecules and the rock surface at elevated pressures, which increases fluid density and promotes stronger wetting tendencies (Alanazi et al., 2025).

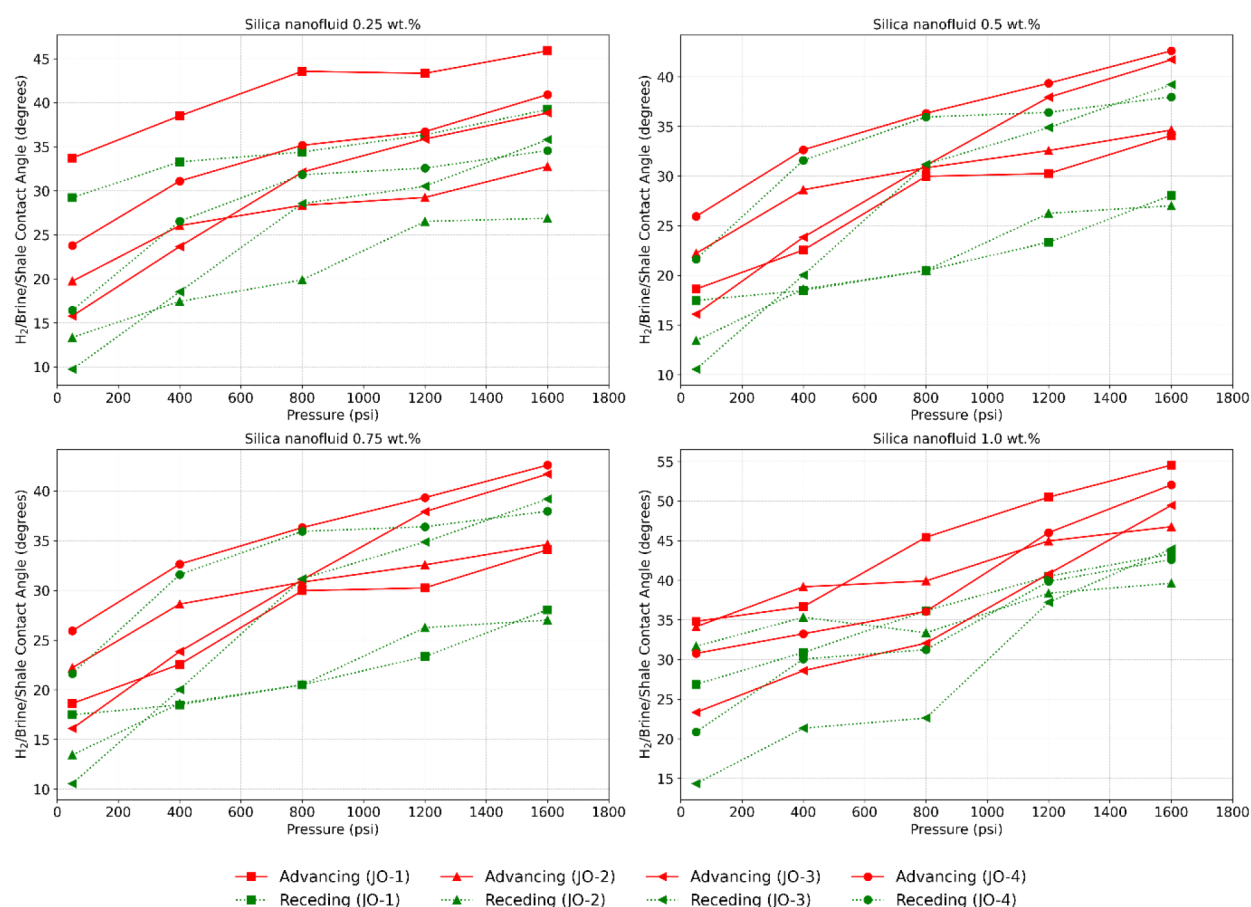


Figure 8—Sensitivity analysis results using trained ML model.

The close alignment between predicted and expected trends indicates that the XGBoost model effectively captures the complex interplay between pressure, nanofluid concentration, and wettability. This underscores the robustness of the model for simulating wettability behavior in hydrogen storage applications, where fluid-rock interactions significantly affect storage efficiency and flow dynamics. Moreover, the model proves particularly useful in scenarios where experimental data is limited or unavailable. For example, at intermediate concentrations such as 0.25, 0.5, and 0.75 wt.%, where experimental testing was not conducted, the model delivers consistent and reliable predictions based on learned trends. This ability to generalize unseen conditions positions the model as a powerful predictive tool for early-stage research, design optimization, and extrapolation beyond experimentally tested scenarios.

Conclusions

Organic-rich shale caprocks tend to become hydrogen-wet due to the presence of organic matter, which reduces their sealing capacity for underground hydrogen storage (UHS). Silica nanofluids can effectively restore water-wet conditions, improving containment. However, experimental evaluation of wettability and interfacial tension is time-consuming and costly. This study demonstrates that machine learning (ML) models offer a powerful alternative to laboratory testing for predicting critical wettability parameters. A suite of ML models—gradient boosting, XGBoost, CatBoost, random forest, and extra trees—was applied to predict advancing and receding contact angles using an extensive dataset from lab experiments conducted across a wide range of pressures, temperatures, and silica nanofluid concentrations. Visualization techniques such as swarm maps and heatmaps helped interpret the dataset. Bayesian optimization, k-fold cross-validation, and grid search were used to fine-tune model hyperparameters. The results showed that XGBoost and CatBoost provided highly accurate predictions, with training and testing R^2 values exceeding 0.998 and 0.949, respectively. Model predictions aligned closely with experimental outcomes, confirming their reliability. Feature importance analysis identified silica nanofluid concentration as the dominant factor affecting wettability, particularly receding contact angles. Pressure and temperature were also found to significantly influence wettability and, by extension, the storage potential of hydrogen in geological formations. This study is the first to apply data-driven ML models to predict wettability behavior in silica-treated organic-rich shales. The findings provide a robust and scalable method for optimizing UHS system designs, offering significant advantages over traditional experimental approaches.

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